ENGINEERING MAJORS SURVEY

Design Package

About the Survey

As more students than ever are pursuing and earning an engineering bachelor's degree in the U.S., questions about their post-graduation pathways are increasing. Which skills have they developed? How are they applying these skills? What are their interests, passions, and goals? Are they designing—and leading—technological and social innovations? Which college experiences are associated with their confidence to innovate, their technical accomplishments, their creative breakthroughs?

To address these questions, the Epicenter research team, whose works is collectively known as Fostering Innovative Generations Studies (FIGS), has launched a major longitudinal study of engineering students' interests and career goals surrounding innovation and entrepreneurship (I&E). This study is part of a larger set of research projects being pursued by the FIGS team, ranging from investigations of I&E programs for engineering students, to comparative analyses of engineering and business students' entrepreneurial intent, to explorations of how I&E might be added to traditional engineering coursework.

For this study, a 35-question survey, the "Engineering Majors Survey", was administered to over 30,000 engineering juniors and seniors across a nationally representative sample of 26 U.S. engineering schools in February-March 2015. This survey draws upon psychological theories of career choice to ask students about their "innovation self-efficacy", expectations for the outcomes of innovative behaviors, innovation interests, and goals around doing innovative work in their early careers. This survey also is designed to measure a range of undergraduate learning experiences that may influence students' beliefs about their ability to innovate, and includes measures of students' entrepreneurial activities, past, present, and future. Findings will be shared starting Summer 2015.

In early 2016, a follow-up survey will be administered to Engineering Majors Survey respondents, many of whom will have graduated by that time and entered the workforce. This longitudinal view will show how engineering students' interests and goals surrounding innovation change over time, in concert with new workplace environments and experiences. The FIGS team hopes to continue to administer follow-up surveys throughout the first ten years of respondents' careers. When complete, this will be among the largest and most extensive studies to date of engineering students' innovation pathways, building from a nationally representative sample of engineering schools and theory-driven instrument design and analysis.

epicenter National Center for Engineering Pathways to Innovation
NSF
Stanford University
VENTUREWELL

Part 1: Annotated Survey Instrument

This is an annotated version of the Winter/Spring 2015 Engineering Majors Survey instrument showing the sources and reference points for each question, as well as how sections and select items fit into the Social Cognitive Career Theory (SCCT) framework (see Lent, Brown, & Hackett, 1994). Annotation is in red font below each survey question. For some survey questions, red numbers have been added to each item to facilitate explanation.

The formatting of this questionnaire on paper does not match its online formatting in Limesurvey, but basic features should be apparent. This document includes logic for survey branching where relevant.

Overall, the survey's design was informed by Lent and Brown's (2006) "measurement guide" for SCCT constructs and other research on questionnaire development (e.g., Krosnick & Presser, 2010). For more information about the survey project, including the institutional sampling method, please see http://epicenter.stanford.edu/page/engineering-majors-survey .

This document is accompanied by a reference list that contains complete references for all papers cited, and a Power Point slide that maps SCCT domains to survey questions and shows hypothesized relationships between constructs. The annotated instrument is separate from a codebook for the final dataset; this codebook will be developed in Spring/Summer 2015.

This survey was conducted with support from the National Center for Engineering Pathways to Innovation (Epicenter), a center funded by the National Science Foundation (grant number DUE-1125457) and directed by Stanford University and VenureWell, formerly the National Collegiate Inventors and Innovators Alliance (NCIIA). Suggested citation for this document: National Center for Engineering Pathways to Innovation (Epicenter). 2015. The Engineering Majors Survey Annotated Instrument.

The Engineering Majors Survey

Your experiences are important to the future of engineering education. So thanks for taking this survey sponsored by [name of Engineering Majors Survey school] and the National Science Foundation through Epicenter. Your thoughtful responses are greatly appreciated. This survey has 35 questions and is designed to be completed in 10 minutes.

The survey has 5 main sections. We want to know about your:

1. current plan of study
2. school experiences
3. beliefs, expectations, and interests
4. future career goals
5. background

Responses to most questions are mandatory in order to proceed through the survey, but don't worry, you can mark "I prefer not to answer" to any question you don't want to answer. Mandatory questions have a red **[black in this copy]** asterisk.

ENTER TO WIN AN AMAZON GIFT CARD! At the end of the survey, we'll tell you how you can enter a drawing to win one of 200 $25.00 gift cards from Amazon. You'll have a chance to provide your permanent email address so that we can stay in touch with you.

Ready to go? Please click on the "Next" button below to continue.

Research Information and Consent

Please read the information below and confirm your consent to participate.

STUDY TITLE: Fostering Innovative Generations Study (FIGS)

DESCRIPTION: You are invited to participate in a research study funded by the National Science Foundation through Epicenter aimed at understanding what kinds of experiences contribute to engineering students' current and post-graduation interests and career goals, particularly around innovation and entrepreneurship. This study will take place in winter 2015. To participate in the study, you must be at least 18 years of age and intend to major in an engineering field.

You will be asked to complete an internet-based questionnaire where you will be asked to provide some information about your undergraduate coursework and extracurricular activities, and future career plans. The survey will take approximately 10 minutes to complete. All information collected will be confidential, and accessed only by authorized researchers who have signed a confidentiality agreement. A short follow-up survey will be conducted in 2016 and you will be invited to provide your permanent email address so that we may invite you to participate. Permanent email addresses will be excluded from any analysis and reporting of the data, and will be strictly

You can download a copy of this information here.

[]I have read the information above and would like to take the survey *

Please choose **only one** of the following:

- Yes
- No

--Respondents who marked "yes" proceeded to page 1 of the survey. Respondents who marked "no" were directed to the concluding thank-you page.

SECTION 1: CURRENT PLAN OF STUDY

In this section, we are interested in learning more about your enrollment status and degree plans.

--This section is intended to provide both background and context measures as part of SCCT.

[]1. What institution are you currently attending? *

Please choose **only one** of the following:

- [insert name of Engineering Majors Survey school]
- Other (please specify):
- I prefer not to answer

--Item adapted from APPLES (Academic Pathways of People Learning Engineering Survey) (2008). .

[]2. What is your current academic standing? *

Please choose **only one** of the following:

- Freshman
- Sophomore
- Junior
- Senior
- Fifth-year senior or more
- Other (please specify):
- I prefer not to answer

--Item adapted from APPLES (2008).

[]3. Are you enrolled primarily as a: *

Please choose **only one** of the following:

- Full-time student
- Part-time student
- I prefer not to answer

--Item adapted from APPLES (2008).

[]4. Did you transfer to your current institution from another college/university? *

Please choose **only one** of the following:

- No
- Yes
- I prefer not to answer

--Item developed for the Engineering Majors Survey (EMS).

[]5. What is your <u>primary major field of study?</u> *

Please choose **only one** of the following:

Aerospace/Aeronautical/Astronautical Engineering
Agricultural Engineering
Architectural Engineering
Bioengineering/Biomedical Engineering
Biological Engineering
Chemical Engineering
Civil Engineering
Computer Engineering
Computer Engineering Technology
Computer Science
Construction Engineering/Construction Management
Construction Engineering Technology
Electrical/Electronics/Communications Engineering
Electrical Engineering Technology
Electromechanical Engineering Technology
Engineering, general
Engineering Management
Engineering Physics
Engineering Technology
Environmental Engineering
Geological Engineering
Geophysical Engineering
Industrial Engineering
Industrial Engineering Technology
Management Science and Engineering
Manufacturing Engineering
Materials Engineering/Materials Science and Engineering
Mechanical Engineering
Mechanical Engineering Technology
Metallurgical Engineering
Mining Engineering
Nuclear Engineering
Ocean Engineering
Optical Science and Engineering
Petroleum Engineering
Robotics Engineering
Software Engineering
Systems Engineering
Other (please specify):

I prefer not to answer

--List of majors developed from two sources: 1) the Pathways of Engineering Alumni Research Survey (PEARS) (2011); 2) research on major offerings at the EMS schools (majors that were offered by one institution only are not included on the list, with the exception of engineering technology majors).

[]5a. In this major, are you pursuing a: *

Only answer this question if the following conditions are met:
Answer was 'Other' *or* 'Systems Engineering' *or* 'Software Engineering' *or* 'Robotics Engineering' *or* 'Petroleum Engineering' *or* 'Optical Science and Engineering' *or* 'Ocean Engineering' *or* 'Nuclear Engineering' *or* 'Mining Engineering' *or* 'Metallurgical Engineering' *or* 'Mechanical Engineering Technology' *or* 'Mechanical Engineering' *or* 'Materials Engineering/Materials Science and Engineering' *or* 'Manufacturing Engineering' *or* 'Management Science and Engineering' *or* 'Industrial Engineering Technology' *or* 'Industrial Engineering' *or* 'Geophysical Engineering' *or* 'Geological Engineering' *or* 'Environmental Engineering' *or* 'Engineering Technology' *or* 'Engineering Physics' *or* 'Engineering Management' *or* 'Engineering, general ' *or* 'Electromechanical Engineering Technology' *or* 'Electrical Engineering Technology' *or* 'Electrical/Electronics/Communications Engineering' *or* 'Construction Engineering Technology' *or* 'Construction Engineering/Construction Management' *or* 'Computer Science' *or* 'Computer Engineering Technology' *or* 'Computer Engineering' *or* 'Civil Engineering' *or* 'Chemical Engineering' *or* 'Biological Engineering' *or* 'Bioengineering/Biomedical Engineering' *or* 'Architectural Engineering' *or* 'Agricultural Engineering' *or* 'Aerospace/Aeronautical/Astronautical Engineering' at [S1Q5]' (5. What is your primary major field of study?)

Please choose **only one** of the following:

- Bachelor of science (BS) degree
- Bachelor of arts (BA) degree
- I prefer not to answer

--Item developed for EMS.

[]5b. Are you pursuing a concentration or topical track within this major? *

Only answer this question if the following conditions are met:
Answer was 'Other' *or* 'Systems Engineering' *or* 'Software Engineering' *or* 'Robotics Engineering' *or* 'Petroleum Engineering' *or* 'Optical Science and Engineering' *or* 'Ocean Engineering' *or* 'Nuclear Engineering' *or* 'Mining Engineering' *or* 'Metallurgical Engineering' *or* 'Mechanical Engineering Technology' *or* 'Mechanical Engineering' *or* 'Materials Engineering/Materials Science and Engineering' *or* 'Manufacturing Engineering' *or* 'Management Science and Engineering' *or* 'Industrial Engineering Technology' *or* 'Industrial Engineering' *or* 'Geophysical Engineering' *or* 'Geological Engineering' *or* 'Environmental Engineering' *or* 'Engineering Technology' *or* 'Engineering Physics' *or* 'Engineering Management' *or* 'Engineering, general ' *or* 'Electromechanical Engineering Technology' *or* 'Electrical Engineering Technology' *or* 'Electrical/Electronics/Communications Engineering' *or* 'Construction Engineering Technology' *or* 'Construction Engineering/Construction Management' *or* 'Computer Science' *or* 'Computer Engineering Technology' *or* 'Computer Engineering' *or* 'Civil Engineering' *or* 'Chemical Engineering' *or* 'Biological Engineering' *or* 'Bioengineering/Biomedical Engineering' *or* 'Architectural Engineering' *or* 'Agricultural Engineering' *or* 'Aerospace/Aeronautical/Astronautical Engineering' at [S1Q5]' (5. What is your primary major field of study?)

Please choose **only one** of the following:

- No
- Yes
- I prefer not to answer

--Item developed for EMS.

[]What is the concentration or topical track in which you are enrolled? (Please enter the name of your concentration or topical track area in the space provided.)

Only answer this question if the following conditions are met:
Answer was 'Yes' at [S1Q5b]' (5b. Are you pursuing a concentration or topical track within this major?)

Please write your answer(s) here:

Name of concentration/topical track: ____________________

--Item developed for EMS.

[]5c. Do you intend to complete this major? *

Only answer this question if the following conditions are met:
Answer was 'Other' *or* 'Systems Engineering' *or* 'Software Engineering' *or* 'Robotics Engineering' *or* 'Petroleum Engineering' *or* 'Optical Science and Engineering' *or* 'Ocean Engineering' *or* 'Nuclear Engineering' *or* 'Mining Engineering' *or* 'Metallurgical Engineering' *or* 'Mechanical Engineering Technology' *or* 'Mechanical Engineering' *or* 'Materials Engineering/Materials Science and Engineering' *or* 'Manufacturing Engineering' *or* 'Management Science and Engineering' *or* 'Industrial Engineering Technology' *or* 'Industrial Engineering' *or* 'Geophysical Engineering' *or* 'Geological Engineering' *or* 'Environmental Engineering' *or* 'Engineering Technology' *or* 'Engineering Physics' *or* 'Engineering Management' *or* 'Engineering, general ' *or*

'Electromechanical Engineering Technology' *or* 'Electrical Engineering Technology' *or* 'Electrical/Electronics/Communications Engineering' *or* 'Construction Engineering Technology' *or* 'Construction Engineering/Construction Management' *or* 'Computer Science' *or* 'Computer Engineering Technology' *or* 'Computer Engineering' *or* 'Civil Engineering' *or* 'Chemical Engineering' *or* 'Biological Engineering' *or* 'Bioengineering/Biomedical Engineering' *or* 'Architectural Engineering' *or* 'Agricultural Engineering' *or* 'Aerospace/Aeronautical/Astronautical Engineering' at [S1Q5]' (5. What is your primary major field of study?)

Please choose **only one** of the following:

- Definitely will not
- Probably will not
- Might or might not
- Probably will
- Definitely will
- I prefer not to answer

--Item adapted from APPLES (2008).

[]6. Are you pursuing a second (double) major? *

Please choose **only one** of the following:

- No
- Yes
- I prefer not to answer

--Item developed for EMS.

[]6a. In which field of study? *

Only answer this question if the following conditions are met:
Answer was 'Yes' at [S1Q6]' (6. Are you pursuing a second (double) major?)

Please choose **only one** of the following:

Aerospace/Aeronautical/Astronautical Engineering

Agricultural Engineering

Architectural Engineering

Bioengineering/Biomedical Engineering

Biological Engineering

Chemical Engineering

Civil Engineering

Computer Engineering

Computer Engineering Technology

Computer Science

Construction Engineering/Construction Management

Construction Engineering Technology

Electrical/Electronics/Communications Engineering

Electrical Engineering Technology

Electromechanical Engineering Technology

Engineering, general

Engineering Management

Engineering Physics

Engineering Technology

Environmental Engineering

Geological Engineering

Geophysical Engineering

Industrial Engineering

Industrial Engineering Technology

Management Science and Engineering

Manufacturing Engineering

Materials Engineering/Materials Science and Engineering

Mechanical Engineering

Mechanical Engineering Technology

Metallurgical Engineering

Mining Engineering

Nuclear Engineering

Ocean Engineering

Optical Science and Engineering

Petroleum Engineering

Robotics Engineering

Software Engineering

Systems Engineering

Other (please specify):

I prefer not to answer

--List of majors developed from two sources: 1) PEARS (2011); 2) research on major offerings at the EMS schools (majors that were offered by one institution only are not included on the list, with the exception of engineering technology majors).

[]6b. In this major, are you pursuing a: *

Only answer this question if the following conditions are met:
Answer was 'Other' *or* 'Systems Engineering' *or* 'Software Engineering' *or* 'Robotics Engineering' *or* 'Petroleum Engineering' *or* 'Optical Science and Engineering' *or* 'Ocean Engineering' *or* 'Nuclear Engineering' *or* 'Mining Engineering' *or* 'Metallurgical Engineering' *or* 'Mechanical Engineering Technology' *or* 'Mechanical Engineering' *or* 'Materials Engineering/Materials Science and Engineering' *or* 'Manufacturing Engineering' *or* 'Management Science and Engineering' *or* 'Industrial Engineering Technology' *or* 'Industrial Engineering' *or* 'Geophysical Engineering' *or* 'Geological Engineering' *or* 'Environmental Engineering' *or* 'Engineering Technology' *or* 'Engineering Physics' *or* 'Engineering Management' *or* 'Engineering, general ' *or* 'Electromechanical Engineering Technology' *or* 'Electrical Engineering Technology' *or* 'Electrical/Electronics/Communications Engineering' *or* 'Construction Engineering Technology' *or* 'Construction Engineering/Construction Management' *or* 'Computer Science' *or* 'Computer Engineering Technology' *or* 'Computer Engineering' *or* 'Civil Engineering' *or* 'Chemical Engineering' *or* 'Biological Engineering' *or* 'Bioengineering/Biomedical Engineering' *or* 'Architectural Engineering' *or* 'Agricultural Engineering' *or* 'Aerospace/Aeronautical/Astronautical Engineering' at [S1Q6a]' (6a. In which field of study?)

Please choose **only one** of the following:

- Bachelor of science (BS) degree
- Bachelor of arts (BA) degree
- I prefer not to answer

--Item developed for EMS.

[]6c. Are you pursuing a concentration or topical track within this major? *

Only answer this question if the following conditions are met:
Answer was 'Other' *or* 'Systems Engineering' *or* 'Software Engineering' *or* 'Robotics Engineering' *or* 'Petroleum Engineering' *or* 'Optical Science and Engineering' *or* 'Ocean Engineering' *or* 'Nuclear Engineering' *or* 'Mining Engineering' *or* 'Metallurgical Engineering' *or* 'Mechanical Engineering Technology' *or* 'Mechanical Engineering' *or* 'Materials Engineering/Materials Science and Engineering' *or* 'Manufacturing Engineering' *or* 'Management Science and Engineering' *or* 'Industrial Engineering Technology' *or* 'Industrial Engineering' *or* 'Geophysical Engineering' *or* 'Geological Engineering' *or* 'Environmental Engineering' *or* 'Engineering Technology' *or* 'Engineering Physics' *or* 'Engineering Management' *or* 'Engineering, general ' *or* 'Electromechanical Engineering Technology' *or* 'Electrical Engineering Technology' *or* 'Electrical/Electronics/Communications Engineering' *or* 'Construction Engineering Technology' *or* 'Construction Engineering/Construction Management' *or* 'Computer Science' *or* 'Computer Engineering Technology' *or* 'Computer Engineering' *or* 'Civil Engineering' *or* 'Chemical Engineering' *or* 'Biological Engineering' *or* 'Bioengineering/Biomedical Engineering' *or* 'Architectural Engineering' *or* 'Agricultural Engineering' *or* 'Aerospace/Aeronautical/Astronautical Engineering' at [S1Q6a]' (6a. In which field of study?)

Please choose **only one** of the following:

- No
- Yes
- I prefer not to answer

--Item developed for EMS.

[]What is the concentration or topical track in which you are enrolled? (Please enter the name of your concentration or topical track area in the space provided.)

Only answer this question if the following conditions are met:
Answer was 'Yes' at [S1Q6c]' (6c. Are you pursuing a concentration or topical track within this major?)

Please write your answer(s) here:

Name of concentration/topical track: ____________________

--Item developed for EMS.

[]6d. Do you intend to complete this second major? *

Only answer this question if the following conditions are met:
Answer was 'Other' *or* 'Systems Engineering' *or* 'Software Engineering' *or* 'Robotics Engineering' *or* 'Petroleum Engineering' *or* 'Optical Science and Engineering' *or* 'Ocean Engineering' *or* 'Nuclear Engineering' *or* 'Mining Engineering' *or* 'Metallurgical Engineering' *or* 'Mechanical Engineering Technology' *or* 'Mechanical Engineering' *or* 'Materials Engineering/Materials Science and Engineering' *or* 'Manufacturing Engineering' *or* 'Management Science and Engineering' *or* 'Industrial Engineering Technology' *or* 'Industrial Engineering' *or* 'Geophysical Engineering' *or* 'Geological Engineering' *or* 'Environmental Engineering' *or* 'Engineering Technology' *or* 'Engineering Physics' *or* 'Engineering Management' *or* 'Engineering, general ' *or* 'Electromechanical Engineering Technology' *or* 'Electrical Engineering Technology' *or* 'Electrical/Electronics/Communications Engineering' *or* 'Construction Engineering Technology' *or* 'Construction Engineering/Construction Management' *or* 'Computer Science' *or* 'Computer Engineering Technology' *or* 'Computer Engineering' *or* 'Civil Engineering' *or* 'Chemical Engineering' *or* 'Biological Engineering' *or* 'Bioengineering/Biomedical Engineering' *or* 'Architectural Engineering' *or* 'Agricultural Engineering' *or* 'Aerospace/Aeronautical/Astronautical Engineering' at [S1Q6a]' (6a. In which field of study?)

Please choose **only one** of the following:

- Definitely will not
- Probably will not
- Might or might not
- Probably will
- Definitely will
- I prefer not to answer

--Item adapted from APPLES (2008).

SECTION 1: CURRENT PLAN OF STUDY

(Section 1 continued)

In this section, we are interested in learning more about your enrollment status and degree plans.

[]7. In addition to your major(s), are you pursuing a minor or certificate for academic credit? *

Please choose **all** that apply:

- ☐ No
- ☐ Yes, a minor
- ☐ Yes, a certificate
- ☐ I prefer not to answer

--Item developed for EMS and informed by Gilmartin et al. (2014) and Shartrand et al. (2010).

[]What is the minor program in which you are enrolled?

Only answer this question if the following conditions are met:
Answer was 'Yes, a minor' at
[S1Q7]' (7. In addition to your major(s), are you pursuing a minor or certificate for academic credit?)

Please write your answer(s) here:

Name of program:

--Item developed for EMS.

[]What is the certificate program in which you are enrolled?

Only answer this question if the following conditions are met:
Answer was 'Yes, a certificate' at [S1Q7]' (7. In addition to your major(s), are you pursuing a minor or certificate for academic credit?)

Please write your answer(s) here:

Name of program: __________

--Item developed for EMS.

[]8. During which year do you anticipate graduating from your current institution with your bachelor's degree(s)? *

Please choose **only one** of the following:

- ○ 2015
- ○ 2016
- ○ 2017
- ○ 2018
- ○ 2019 or later
- ○ I plan to transfer to another institution before graduating
- ○ I plan to leave my institution before graduating and do not have plans to re-enroll elsewhere
- ○ I do not know
- ○ I prefer not to answer

--Item developed for EMS.

SECTION 2. SCHOOL EXPERIENCES

--This section is intended to provide learning experience and context measures as part of SCCT.

In this section, we would like to know more about your learning experiences and activities during high school and as an undergraduate student, especially as these relate to "**innovation**" and "**entrepreneurship**".

[]9. <u>During high school,</u> did you: *

Please choose the appropriate response for each item:

	No	Yes	I prefer not to answer
1 Take an art, dance, music, theater, or creative writing class			
2 Learn computer programming			
3 Take a shop class (e.g., a woodworking, automotive, or maker class) or engineering class			
4 Participate in a robotics competition, such as a FIRST Robotics Competition			
5 Attend a science, math, technology, or engineering related summer camp			
6 Have a research position or internship at a science, math, technology, or engineering related company or organization			
7 Learn about entrepreneurship			
8 Start or co-found your own club, organization, or company			

--Items developed for EMS. Last item informed by PEARS (2011). These items were designed to coordinate with those in Q11 and Q12, i.e., serve as high school analogs to the college-level experiences that may influence students' innovation self-efficacy and innovation outcome expectations (see Section 3).

[]10. <u>While an undergraduate</u>, have you done (or are you currently doing) each of the following for at least one full academic or summer term? *

Please choose the appropriate response for each item:

	No	Yes	I prefer not to answer
1 Conduct research with a faculty member			
2 Work in a professional engineering environment as an intern/co-op			
3 Have a work-study or other type of job to help pay for your college education			
4 Participate in study abroad			

--Item 1 adapted and modified from APPLES (2008). Items 2 and 4 adapted from PEARS (2011). Item 3 developed for EMS. These items are non-innovation-specific learning experiences that may differentiate engineering student experiences from those of their peers, and/or have some association with important engineering career-related outcomes (see, for example, Brunhaver, forthcoming, Brunhaver et al., 2012, Lichtenstein et al., 2010, Sheppard et al., 2010). Put differently, these items could reflect educational contexts and experiences that matter to broader pathway decisions of engineers (in a more general SCCT model). Item 3 also is included as a way to triangulate findings related to students' socioeconomic status (SES) (see Section 5).

[]11. <u>As part of your undergraduate coursework so far (including courses you are currently taking),</u> have you taken courses that include any of the following topics or components? *

Please choose the appropriate response for each item:

	No	Yes	I don't know	I prefer not to answer
1 Art, dance, music, theater, or creative writing				
2 Computer science				

3 Theory of design
4 Designing and/or prototyping things or ideas
5 Business or enterprise topics (including entrepreneurship or venture creation)
6 Leadership topics
7 Interaction with students from non-engineering majors

--Items developed for EMS. These items are designed to work in conjunction with those in Q9 and Q12, as curricular-based college learning experiences that may influence students' innovation self-efficacy/innovation outcome expectations (Q12 measures co- and extra-curricular college learning experiences; Q9 measures high school learning experiences). Items 5 and 6 are informed by the Fostering Innovative Generations Studies (FIGS) examination of entrepreneurship programs for engineers (Gilmartin et al., 2014).

[]12. Below are various extra- and co-curricular activities you may have been involved in while an undergraduate. Many of these have to do with innovation and/or entrepreneurship; others are more general to the college experience. <u>Please mark which of the following you have done during your undergraduate years so far (including activities you are currently doing).</u> *

Please choose **all** that apply:

Participated in...

☐ 1 A business or entrepreneurship club

☐ 2 A community service-based club (e.g., Engineers Without Borders, Design for America, EPICS)

☐ 3 A design club

☐ 4 A robotics club

☐ 5 Other student clubs or groups in engineering

☐ 6 Other student clubs or groups outside of engineering

Entered...

☐ 7 A business plan, business model, or elevator pitch competition

☐ 8 A design or invention competition

☐ 9 A social entrepreneurship/social innovation competition (e.g., the Dell Social Innovation Challenge)

Made use of...

☐ 10 A maker space/design or inventors studio/prototyping lab

Attended...

☐ 11 A career related event or meeting (e.g., a college career fair, a one-on-one meeting with a career counselor)

☐ 12 A speaker series or related presentations about entrepreneurship and/or innovation

☐ 13 A start-up bootcamp (e.g., Start-up Weekend, 3-Day Startup)

☐ 14 A presentation on a new engineering technology, process, or design (outside of class)

Lived in...

☐ 15 A residential or dorm-based <u>engineering</u> program/<u>engineering</u> living-learning community

☐ 16 A residential or dorm-based <u>entrepreneurship or innovation</u> program/<u>entrepreneurship or innovation</u> living learning community

Received...

☐ 17 Funding from a program to finance new ideas

Led...

☐ 18 A student organization

Started or co-founded...

☐ 19 A student club or other student group on campus

☐ 20 Your own for-profit or non-profit organization

☐ 21 None of the above (I have not done any of the activities in this question)

☐ 22 I prefer not to answer

-- These items represent a mix of EMS-specific measures and measures adapted from or informed by other instruments. Some were developed to align with activities included on Epicenter's University Innovation Fellows (UIF) landscape inventories and/or activities measured in other Epicenter data efforts (e.g., Items 7-10, 12-13 and 16-17) (see Epicenter , 2015a and Epicenter, 2015b, as well as http://universityinnovation.org/wiki/How_to_complete_the_Landscape_Canvas). The first six items were designed to elaborate on two more general items in APPLES (2008) (see Sheppard et al., 2010). Items 19 and 20 were designed to elaborate on PEARS (2011). Items 11, 14, 15, and 18 were developed for EMS. Intended to work with Q9 and Q11, these items are college-level co- and extra-curricular experiences that may influence students' innovation self-efficacy and innovation outcome expectations.

[]13. In the past year, how often have you discussed each of the following with faculty members at your institution? *

Please choose the appropriate response for each item:	Never	Rarely	Sometimes	Often	Very often	I prefer not to answer
1 Course topics and assignments (not during class or section time)						
2 Your professional options with an engineering degree						
3 New design or business ideas						

--Items 1 and 2 adapted from PEARS (2011). Item 3 developed for EMS. These items may reflect learning experiences and social contexts.

[]14. In the past year, how often have you discussed each of the following with other students? *

Please choose the appropriate response for each item:

	Never	Rarely	Sometimes	Often	Very often	I prefer not to answe
1 Course topics and assignments (not during class or section time)						
2 Your professional options with an engineering degree						
3 New design or business ideas						

--Items 1 and 2 adapted from PEARS (2011). Item 3 developed for EMS. These items may reflect learning experiences and social contexts.

[] The most detailed page of the survey is now behind you! You're making great progress.

SECTION 3. BELIEFS, EXPECTATIONS, AND INTERESTS

--This section includes items that will compose the Innovation Self-Efficacy, Innovation Outcome Expectations, and Innovation Interests scales in the final SCCT model.

In this section, we would like to learn more about your confidence in various skills and abilities, as well as your perceptions of the outcomes of some of these skills and abilities in your future career. The last question in this section asks you about your interests in doing certain activities.

[]15. How confident are you in your ability to do each of the following at this time? *

Please choose the appropriate response for each item:

	Not confident	Slightly confident	Moderately confident	Very confident	Extremely confident	I prefer not to answer
1 Ask a lot of questions						
2 Generate new ideas by observing the world						
3 Experiment as a way to understand how things work						
4 Actively search for new ideas through experimenting						
5 Build a large network of contacts with whom you can interact to get ideas for new products or services						
6 Connect concepts and ideas that appear, at first glance, to be unconnected						
7 Design a new product or project to meet specified requirements						
8 Conduct experiments, build prototypes, or construct mathematical models to develop or evaluate a design						
9 Develop and integrate component sub-systems to build a complete system or product						
10 Analyze the operation or functional performance of a complete system						
11 Troubleshoot a failure of a technical component or system						
12 Lead a team of people						
13 Communicate your ideas effectively to people in different positions or fields						
14 Take the steps needed to place a financial value on a new business venture						

--The first six items, in blue, are the items designed to compose the Innovation Self-Efficacy scale. These items were adapted and modified from Dyer et al. (2008) (with Dyer's permission). A larger set of such items was piloted and analyzed in November 2014 as part of the EMS development process. These six items were the highest loading items on the factors that emerged in these pilot analyses (factors corresponded to Dyer et al.'s innovative behavior domains of "questioning", "observing", "experimenting", and "idea networking", in addition to their concept of "associative thinking"). As a whole, these six items tend to emphasize the "discovery", or "idea generation", part of innovation (Dyer et al., 2011), which becomes particularly relevant when thinking about correlations with our other "core" SCCT constructs in this survey: Innovation Outcome Expectations, Innovation Interests, and Career Goals: Innovative Work (Qs 16, 17, and 18, respectively).

--The next five items, in green, compose the Engineering Task Self-Efficacy scale, and were adapted and modified from Fouad and Singh (2011) (with Fouad's permission). The ETSA is conceived as a potential additional correlate of Qs 16, 17, and/or 18, as well as other career-related plans measures in Section 4. These items were included on PEARS (2011).

--The next two items, in purple, are adapted from PEARS (2011), and represent select measures of professional/interpersonal self-efficacy (engineers' professional/interpersonal confidence was a central construct in Sheppard et al.'s 2010 analyses of APPLES data). These are experimental items in the context of the proposed SCCT model.

--The last item, in orange, is adapted and modified from Lucas et al.'s (2009) Venturing Self-Efficacy scale. Schar et al. (2014) found a strong correlation between the original version of this item and engineering students' entrepreneurial intent. This is an experimental item in the context of the proposed SCCT model.

--The order of items in this question was randomized for survey respondents.

[]16. Imagine the work you will be doing in the first year after you graduate. Estimate what will happen if you "ask a lot of questions" in this work.

If I ask a lot of questions... *

Please choose the appropriate response for each item:

	Definitely will not	Probably will not	Might or might not	Probably will	Definitely will	I prefer not to answer
1 I will be seen as a "star" in this work.						
2 I will earn the respect of my colleagues.						
3 I will hurt my chances for moving ahead.						
4 I will be seen as a troublemaker.						

-These items are designed to compose the Innovation Outcome Expectations scale. The prompt "ask a lot of questions" is designed to coordinate with the same item in Q15 (as part of the Innovation Self-Efficacy scale); it is, in other words, one of the innovative behavior indicators identified by Dyer et al. (2008) and modified for use in the EMS. The four outcome measures are informed by Fouad and Singh (2011). These four measures may work as two sub-scales (positive OE and negative OE) rather than one single scale pending psychometric analyses.

[]17. How much interest do you have in: *

Please choose the appropriate response for each item:

	Very low interest	Low interest	Medium interest	High interest	Very high interest	I prefer not to answer
1 Experimenting in order to find new ideas						
2 Giving an "elevator pitch" or presentation to a panel of judges about a new product or business idea						
3 Finding resources to bring new ideas to life						
4 Developing plans and schedules to implement new ideas						
5 Conducting basic research on phenomena in order to create new knowledge						
6 Working on products, projects, or services that address societal challenges						
7 Working on products, projects, or services that have significant financial potential						

--These items reflect Innovation Interests, some of which may be analyzed in a scale pending psychometric analyses. The first item is designed to coordinate with a similarly worded item in Q15 (as part of the Innovation Self-Efficacy scale), the prompt in Q16 (also part of the Innovation Self-Efficacy scale in Q15), and the first two items in Q18, Career Goals: Innovative Work (all of these items tend to emphasize discovery or idea-generation aspects of innovation). The next three items are designed to coordinate with the post-idea-generation stages of innovation captured in Items 3, 4, and 5 in Q18; Item 2 is adapted from Duval-Couetil et al. (2012) (with Duval-Couetil's permission). Item 5 was developed for the EMS as a way to highlight interests in academic or scholarly innovation (a measure of scholarly innovation was included on the Stanford Innovation Survey, 2011; see Eesley & Miller, 2012); Items 6 and 7 are inspired by Lintl et al. (2015) and the YES project (2012-present), as well as Gilmartin et al.'s discussion of entrepreneurship programs that include social entrepreneurship as part of their mission (manuscript in preparation).

--The response scale for this question is adapted from the instrument used in Lent et al. (2005).

SECTION 4. FUTURE CAREER GOALS

--This section includes items that will compose the Career Goals: Innovative Work scale—the proposed dependent variable—in the final SCCT model. This section also includes questions about career plans that can help to clarify and deepen understanding of the dependent variable.

This section includes a series of questions about your plans and goals after you graduate.

[]18. How important is it to you to be involved in the following job or work activities in the <u>first five years after you graduate</u>? *

Please choose the appropriate response for each item:

	Not important	Slightly important	Moderately important	Very important	Extremely important	I prefer not to answe
1 Searching out new technologies, processes, techniques, and/or product ideas						
2 Generating creative ideas						
3 Promoting and championing ideas to others						
4 Investigating and securing resources needed to implement new ideas						
5 Developing adequate plans and schedules for the implementation of new ideas						
6 Selling a product or service in the marketplace						

--The items in this question are designed to compose the Career Goals: Innovative Work scale, which serves as the dependent variable in the proposed SCCT model. The primary source for these items come from Kanter's (1988) examination of social and structural conditions that interact with the innovation process—namely, the conditions that catalyze (or not) the four major innovation tasks of "idea generation", "coalition building", "idea realization", and "transfer/diffusion". In Kanter's work, these four micro-level tasks are accomplished by individuals or groups of individuals within organizations, and are shaped and mediated by more macro-level social and structural forces.

Later, Scott and Bruce (1994) and then the YES survey instrument (YES, 2012-present) operationalized these four tasks as innovation behaviors (Items 1-5 in Q18, where—we propose—the first two items reflect "idea generation", the third reflects "coalition-building", the fourth reflects "idea realization", and the fifth reflects both "idea realization" and "transfer/diffusion") (we did not find information as to the exact mapping of item to innovation task in the description of the original scale development, so our map is an inference). These previous studies measured the frequency with which individuals engaged in these behaviors (labeling their scale "Innovation Orientation") (Jin et al., 2014, presents an example of how the Innovation Orientation scale works with engineering students). For the purposes of our work, we modified these items such that they are measured on an importance scale, rather than a frequency scale, to address the idea of "Career Goals" in SCCT. We designed and added Item 6 to the question as a way to further specify Kanter's "transfer/diffusion" innovation task.

Please choose **only one** of the following:

Definitely will not

Probably will not

Might or might not

Probably will

Definitely will

I prefer not to answer

[]19a. If you do enter graduate school in the first five years after you graduate, which degree(s) will you pursue?

Degree 1: Type *

Only answer this question if the following conditions are met:
Answer was 'Definitely will' *or* 'Probably will' *or* 'Might or might not' *or* 'Probably will not' at [S4Q19]' (19. How likely is it that you will enter graduate school in the first years after you graduate?)

Please choose **only one** of the following:

MA/MS

PhD

MBA

JD (Law)

MD

Other

I don't know

I prefer not to answer

[]Which other degree type?

Only answer this question if the following conditions are met:
Answer was 'Other' at [S4Q19a]' (19a. If you do enter graduate school in the first five years after you graduate, which degree(s) will you pursue? Degree 1: Type)

Please write your answer here:

[]Degree 1: Field *

Only answer this question if the following conditions are met:
Answer was 'Probably will' *or*
'Definitely will' *or* 'Might or might not' *or* 'Probably will not' at [S4Q19]' (19. How likely is it that you will enter graduate school in the first five years after you graduate?) *and* Answer was 'PhD' *or* 'MA/MS' *or* 'Other' *or* '*I prefer not to answer' *or* 'I don't know' at [S4Q19a]' (19a. If you do enter graduate school in the first five years after you graduate, which degree(s) will you pursue? Degree 1: Type)

Please choose **only one** of the following:

Agricultural Business and Production

Agricultural Sciences

Architectural/Environmental Design

Biological/Life Sciences

Business Administration/Management

Communication

Computer and Information Sciences

Conservation and Natural Resources

Criminal Justice/Protective Services

Dentistry

Education

Engineering

Engineering Education

Engineering-Related Technologies

Languages, Linguistics, Literature/Letters

Health and Related Sciences

Home Economics

Law/Prelaw/Legal Studies

Liberal Arts/General Studies

Library Science

Mathematics and Statistics

Medicine

Parks, Recreation, Leisure, and Fitness Studies

Philosophy, Religion, Theology

Physical Sciences

Psychology

Public Affairs

Social Work

Social Sciences and History

Veterinary Medicine

Visual and Performing Arts

Other

I don't know

I prefer not to answer

[]Which other degree field?

Only answer this question if the following conditions are met:
Answer was 'Other' at [S4Q19a2]' (Degree 1: Field)

Please write your answer here:

[]In which engineering field? *

Only answer this question if the following conditions are met:
Answer was 'Engineering' at
[S4Q19a2]' (Degree 1: Field)

Please choose **only one** of the following:

- Aerospace, aeronautical, astronautical engineering
- Agricultural engineering
- Architectural engineering
- Bioengineering and biomedical engineering
- Chemical engineering
- Civil engineering
- Computer and systems engineering
- Construction engineering
- Electrical/electronics/communications engineering
- Engineering sciences, mechanics and physics
- Environmental engineering
- Engineering, general
- Geophysical and geological engineering
- Industrial and manufacturing engineering
- Management science and engineering
- Materials engineering, including ceramics and textiles
- Mechanical engineering
- Metallurgical engineering
- Mining and minerals engineering
- Naval architecture and marine engineering
- Nuclear engineering
- Petroleum engineering
- Technical communication
- Other engineering
- I prefer not to answer

[]In which engineering-related technologies field? *

Only answer this question if the following conditions are met:
Answer was 'Engineering-Related
Technologies' at [S4Q19a2]' (Degree 1: Field)

Please choose **only one** of the following:

- Electrical and electronics technologies
- Industrial production technologies
- Mechanical engineering-related technologies
- Other engineering-related technologies
- I prefer not to answer

[]In which computer and information sciences field? *

Only answer this question if the following conditions are met:
Answer was 'Computer and
Information Sciences' at [S4Q19a2]' (Degree 1: Field)

Please choose **only one** of the following:

- Computer and information sciences, general
- Computer programming
- Computer science
- Computer systems analysis
- Data processing
- Information services and systems
- Other computer and information sciences
- I prefer not to answer

[]Degree 2 (if applicable): Type

Only answer this question if the following conditions are met:
Answer was 'Definitely will' *or*
'Probably will' *or* 'Might or might not' *or* 'Probably will not' at [S4Q19]' (19. How likely is it that you will enter graduate school in the first five years after you graduate?)

Please choose **only one** of the following:

- MA/MS
- PhD
- MBA
- JD (Law)
- MD
- Other
- I don’t know
- I prefer not to answer

[]Which other degree type?

Only answer this question if the following conditions are met:
Answer was 'Other' at [S4Q19a3]' (Degree 2 (if applicable): Type)

Please write your answer here:

[]Degree 2 (if applicable): Field

Only answer this question if the following conditions are met:
Answer was 'Probably will not' *or*
'Might or might not' *or* 'Probably will' *or* 'Definitely will' at [S4Q19]' (19. How likely is it that you will enter graduate school in the first five years after you graduate?) *and* Answer was 'MA/MS' *or* 'PhD' *or* 'Other' *or* 'I prefer not to answer' *or* 'I don’t know' at [S4Q19a3]' (Degree 2 (if applicable): Type)

Please choose **only one** of the following:

- Agricultural Business and Production
- Agricultural Sciences
- Architectural/Environmental Design
- Biological/Life Sciences
- Business Administration/Management
- Communication
- Computer and Information Sciences
- Conservation and Natural Resources
- Criminal Justice/Protective Services
- Dentistry
- Education
- Engineering
- Engineering Education
- Engineering-Related Technologies
- Languages, Linguistics, Literature/Letters
- Health and Related Sciences
- Home Economics
- Law/Prelaw/Legal Studies
- Liberal Arts/General Studies
- Library Science
- Mathematics and Statistics
- Medicine
- Parks, Recreation, Leisure, and Fitness Studies
- Philosophy, Religion, Theology
- Physical Sciences
- Psychology
- Public Affairs
- Social Work
- Social Sciences and History
- Veterinary Medicine
- Visual and Performing Arts
- Other
- I don’t know
- I prefer not to answer

[]Which other degree field?

Only answer this question if the following conditions are met:
Answer was 'Other' at [S4Q19a4]' (Degree 2 (if applicable): Field)

Please write your answer here:

[]In which engineering field?

Only answer this question if the following conditions are met:
Answer was 'Engineering' at
[S4Q19a4]' (Degree 2 (if applicable): Field)

Please choose **only one** of the following:

- Aerospace, aeronautical, astronautical engineering
- Agricultural engineering
- Architectural engineering
- Bioengineering and biomedical engineering
- Chemical engineering
- Civil engineering
- Computer and systems engineering
- Construction engineering
- Electrical/electronics/communications engineering
- Engineering sciences, mechanics and physics
- Environmental engineering
- Engineering, general
- Geophysical and geological engineering
- Industrial and manufacturing engineering
- Management science and engineering
- Materials engineering, including ceramics and textiles
- Mechanical engineering
- Metallurgical engineering
- Mining and minerals engineering
- Naval architecture and marine engineering
- Nuclear engineering
- Petroleum engineering
- Technical communication
- Other engineering

[]In which engineering-related technologies field?

Only answer this question if the following conditions are met:
Answer was 'Engineering-Related Technologies' at [S4Q19a4]' (Degree 2 (if applicable): Field)

Please choose **only one** of the following:

- Electrical and electronics technologies
- Industrial production technologies
- Mechanical engineering-related technologies
- Other engineering-related technologies

[]In which computer and information sciences field?

Only answer this question if the following conditions are met:
Answer was 'Computer and Information Sciences' at [S4Q19a4]' (Degree 2 (if applicable): Field)

Please choose **only one** of the following:

- Computer and information sciences, general
- Computer programming
- Computer science
- Computer systems analysis
- Data processing
- Information services and systems
- Other computer and information sciences

--The entirety of Q19 is adapted and modified from PEARS (2011), and aligns with the National Survey of Recent College Graduates (NSRCG) (2008).

[]20. How likely is it that you will do each of the following in the <u>first five years after you graduate?</u> *

Please choose the appropriate response for each item:

	Definitely will not	Probably will not	Might or might not	Probably will	Definitely will	I prefer not to answe
1 Work as an employee for a small business or start-up company						
2 Work as an employee for a medium- or large-size business						
3 Work as an employee for a non-profit organization (excluding a school or college/university)						
4 Work as an employee for the government, military, or public agency (excluding a school or college/university)						
5 Work as a teacher or educational professional in a K-12 school						
6 Work as a faculty member or educational professional in a college or university						
7 Found or start your own for-profit organization						
8 Found or start your own non-profit organization						

--Items adapted and modified from the Professional and Entrepreneurial Aims and CHaracteristics of Engineering Students (PEACHES) survey (2012). The last two items are designed to elaborate on Lintl et al. (2015).

[]21. How likely is it that your work will involve engineering (e.g., engineering practice, research, management, or sales) in... *

Please choose the appropriate response for each item:

	Definitely will not	Probably will not	Might or might not	Probably will	Definitely will	I prefer not to answer
The first year after you graduate						
Five years after you graduate						
Ten years after you graduate						

--Items adapted and modified from APPLES (2008).

[]22. We have asked a number of questions about your future plans. If you would like to elaborate on what you are planning to do, in the next five years or beyond, please do so here.

Please write your answer here:

--Item developed for EMS.

[] Now onto the fifth and final section...just a few more questions left!

SECTION 5. BACKGROUND

--This section is intended to provide background measures as part of SCCT. The placement of these items at the end of the survey is intentional, as a way to minimize biases in the survey responses that may ensue if respondents were asked to indicate, for instance, their race or gender at the start of the survey. We drew from Danaher and Crandall (2008), Steele and Aronson (1995), and Stricker and Ward (2004) in making this design decision.

In this final section of the survey, we ask you to provide background information about yourself.

[]23. What is your sex? *

Please choose **only one** of the following:

- ○ Female
- ○ Male
- ○ Other (please specify): ____________
- ○ I prefer not to answer

--Item adapted and modified from APPLES (2008).

[]24. What is your racial or ethnic identification? *

Please choose **all** that apply:

- ☐ American Indian or Alaska Native
- ☐ Asian or Asian American
- ☐ Black or African American
- ☐ Hispanic or Latino/a
- ☐ Native Hawaiian or Pacific Islander
- ☐ White
- ☐ Other (please specify):: ____________
- ☐ I prefer not to answer

--Item adapted from APPLES (2008).

[]25. In which year were you born? *

Please choose **only one** of the following:

- 2000 or later
- 1999
- 1998
- 1997
- 1996
- 1995
- 1994
- 1993
- 1992
- 1991
- 1990
- 1989
- 1988
- 1987
- 1986
- 1985
- 1984
- 1983
- 1982
- 1981
- 1980
- 1979
- 1978
- 1977
- 1976
- 1975
- 1974
- 1973
- 1972
- 1971
- 1970
- 1969
- 1968
- 1967
- 1966
- 1965 or earlier
- I prefer not to answer

--Item adapted and modified from APPLES (2008).

[]26. Are you: *

Please choose **only one** of the following:

- A United States Citizen
- A Permanent Resident of the United States
- Other
- I prefer not to answer

--Item adapted from APPLES (2008).

[]27. Were you born in the U.S.? *

Please choose **only one** of the following:

- No
- Yes
- I prefer not to answer

--Item adapted from APPLES (2008).

[]How many years have you lived in the U.S.? Please specify:

Only answer this question if the following conditions are met:
Answer was 'No' at [S5Q27]' (27. Were you born in the U.S.?)

Only numbers may be entered in this field.

Please write your answer here:

--Item developed for EMS.

[]28. Did one or more of your parents/guardians immigrate to the U.S.? *

Please choose **only one** of the following:

- No
- Yes
- I prefer not to answer

--Item adapted from APPLES (2008).

[]29. When you were growing up, would you describe your family as: *

Please choose **only one** of the following:

- Low income
- Lower-middle income
- Middle income
- Upper-middle income
- High income
- I prefer not to answer

--Item adapted and modified from APPLES (2008) and PEARS (2011).

[]30. Who in your life has had experience with starting their own business or organization? *

Please choose **all** that apply:

- Parents/guardians
- Siblings
- Other relatives
- Other friends or contacts
- None of the above (no one)
- I prefer not to answer

--Item developed for EMS and informed by the Stanford Innovation Survey (2011).

[]Below you are asked to mark the highest level of education attained by the two people most responsible for raising you, "Parent 1" and (if applicable) "Parent 2".

[]31. Whom do you refer to as Parent 1? *

Please choose **only one** of the following:

- Mother
- Father
- Stepmother
- Stepfather
- Grandmother
- Grandfather
- Legal guardian
- Other (please specify): ______
- I prefer not to answer

[]32. How much education did your “Parent 1” complete? *

Please choose **only one** of the following:

- Did not finish high school
- Graduated from high school
- Attended college but did not complete degree
- Completed an Associate degree (AA, AS, etc.)
- Completed a Bachelor degree (BA, BS, etc.)
- Completed a Master degree (MA, MS, MBA, etc.)
- Completed a Doctoral or Professional degree (JD, MD, PhD, etc.)
- Don’t know or not applicable
- I prefer not to answer

[]33. Whom do you refer to as Parent 2? *

Please choose **only one** of the following:

- Mother
- Father
- Stepmother
- Stepfather
- Grandmother
- Grandfather
- Legal guardian
- Other (please specify): ____________
- Not applicable (N/A)
- I prefer not to answer

[]34. How much education did your "Parent 2" complete? *

Please choose **only one** of the following:

- Did not finish high school
- Graduated from high school
- Attended college but did not complete degree
- Completed an Associate degree (AA, AS, etc.)
- Completed a Bachelor degree (BA, BS, etc.)
- Completed a Master degree (MA, MS, MBA, etc.)
- Completed a Doctoral or Professional degree (JD, MD, PhD, etc.)
- Don't know or not applicable
- I prefer not to answer

--Questions 31-34 adapted and modified from the YES survey instrument (YES, 2012-present) (with YES permission).

[]35. What is your overall college grade point average? *

Please choose **only one** of the following:

- A or A+ (i.e., 3.9 or above on a 4.0 scale)
- A- (3.5-3.8)
- B+ (3.2-3.4)
- B (2.9-3.1)
- B- (2.5-2.8)
- C+ (2.2-2.4)
- C (1.9-2.1)
- C- or lower (1.8 or less)
- I prefer not to answer

--Item adapted from APPLES (2008).

SECTION 5. BACKGROUND

(Section 5 continued) (the remainder of the survey primarily collects information for the follow-up study and the gift card drawing)

We are very grateful for your time in completing the survey! The future of engineering education needs your voice. And we hope this provided you with an opportunity to reflect on your life as an engineering student.

To continue this line of research in ground-breaking ways, we would like to be able to check in with you again in about a year to find out more about your interests and plans.

[]Would you be willing to receive an invitation to participate in a short follow-up survey next year? *

Please choose **only one** of the following:

- Yes
- No

[]Please provide your name and a permanent email address to receive this invitation. *

Only answer this question if the following conditions are met:
Answer was 'Yes' at [S5B1]' (Would you be willing to receive an invitation to participate in a short follow-up survey next year?)

Please write your answer(s) here:

First name	
Last name	
Permanent Email Address (e.g., a Gmail address)	
Please re-enter your Permanent Email Address	

CONCLUSION

As a thank you for your participation in this survey, we would like to enter your name into a random drawing for one of 200 $25.00 Amazon gift cards. The drawing will take place 6 weeks after the close of the survey and will be conducted by the Engineering Majors Survey research team.

[]Would you like to enter your name into a random drawing for an Amazon gift card as a thank you for participating in this survey? *

Please choose **only one** of the following:

- Yes
- No

[]Please provide your name and a permanent email address to be entered into the drawing. *

Only answer this question if the following conditions are met:
Answer was 'Yes' at [S6Q1]' (Would you like to enter your name into a random drawing for an Amazon gift card as a thank you for participating in this survey?)

Please write your answer(s) here:

First name	
Last name	
Permanent Email Address (e.g., a Gmail address)	

--If a student provided their email address for the follow-up survey invitation, this email address was auto-populated here.

[]Finally, we know you have spent your time with us taking this survey. And you may need to log off. But if you have another minute, we have a question that can help us to develop this research for future generations of engineers:

To what extent did this survey inspire you to think about your education in new or different ways? Please describe.

--This question was designed to collect information about students' reactions to the survey, as a way to inform ongoing research and future follow-up correspondence.

Please write your answer here:

Thank you very much!
Best wishes,
The Engineering Majors Survey Team

survey@research-epicenter.stanford.edu

Submit your survey.
Thank you for completing this survey.

Part 2: Reference List

Reference List for the Engineering Majors Survey

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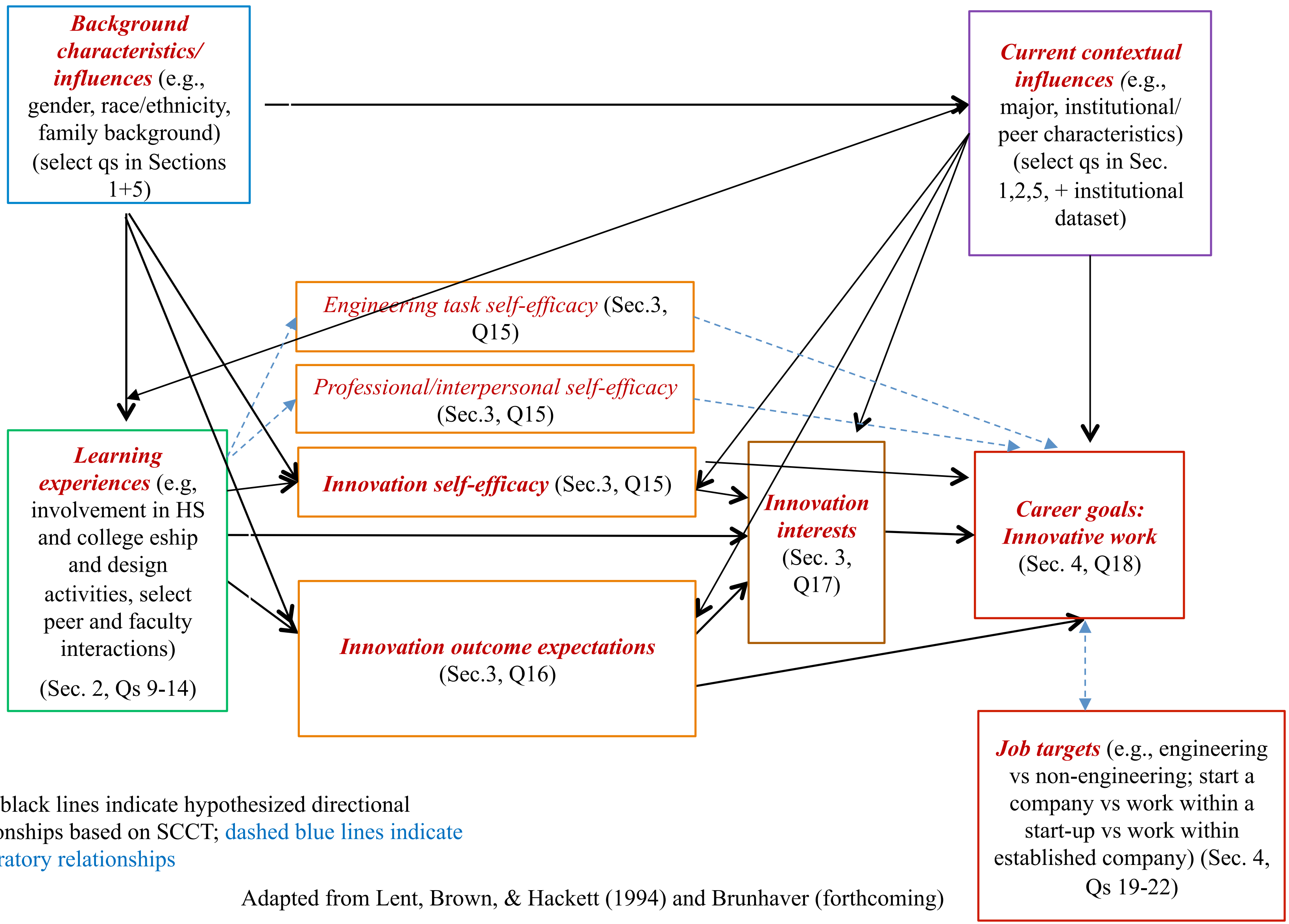
Part 3: Social Cognitive Career Theory

Social Cognitive Career Theory (SCCT) and "The Engineering Majors Survey": Predicting "Career goals: Innovative work" – Time-point 1

--Which survey sections/questions correspond with which SCCT domain(s) are in parentheses in each box



Solid black lines indicate hypothesized directional relationships based on SCCT; dashed blue lines indicate exploratory relationships

Adapted from Lent, Brown, & Hackett (1994) and Brunhaver (forthcoming)